

MIDTERM REPORT

Attendance System Using Face Recognition

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Section D

1. Introduction

Attendance management is a crucial part of any organization, especially in educational institutions. Traditional methods such as manual attendance marking, ID cards, or biometric systems often suffer from issues like proxy attendance, time consumption, and lack of accuracy. To overcome these limitations, modern systems are increasingly adopting artificial intelligence-based solutions.

This project focuses on developing an **automated attendance system using face recognition technology**. The system uses computer vision and machine learning techniques to identify individuals and mark their attendance in real-time. By capturing facial features through a camera and comparing them with stored data, the system ensures accuracy, security, and efficiency.

The main goal of this project is to reduce manual effort, eliminate fraudulent attendance practices, and provide a seamless and contactless solution.

2. Objectives

The primary objectives of this project are:

- To develop a reliable face recognition-based attendance system
 - To automate attendance marking without human intervention
 - To reduce proxy attendance and errors
 - To ensure fast and accurate identification of individuals
 - To maintain attendance records digitally
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3. Problem Statement

Traditional attendance systems face several challenges:

- Manual attendance is time-consuming
- Biometric systems require physical contact
- ID cards can be misused
- Proxy attendance is common
- Data management is inefficient

This project aims to address these issues by implementing a **contactless and automated face recognition system** that improves accuracy and efficiency.

4. Proposed System

The proposed system captures live video input through a camera and detects faces using computer vision techniques. Once a face is detected, it is compared with a trained dataset to identify the person. If the face matches, attendance is automatically marked in the database.

Key Features:

- Real-time face detection
 - Face recognition using trained models
 - Automatic attendance marking
 - Storage of attendance records
 - User-friendly interface
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5. System Architecture

The system consists of the following components:

1. **Camera Module** – Captures real-time video
 2. **Face Detection Module** – Detects faces in the frame
 3. **Face Recognition Module** – Matches faces with stored data
 4. **Database** – Stores user details and attendance records
 5. **User Interface** – Displays results
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6. Methodology

The working of the system is divided into the following steps:

1. Data Collection

Images of individuals are collected and stored in a dataset. These images are used to train the system.

2. Preprocessing

The images are converted to grayscale and resized for better performance.

3. Face Detection

Using algorithms like Haar Cascade, faces are detected from the image or video stream.

4. Model Training

The system is trained using machine learning models such as LBPH (Local Binary Pattern Histogram).

5. Face Recognition

The detected face is compared with stored data to identify the person.

6. Attendance Marking

If a match is found, attendance is marked with date and time.

7. Tools and Technologies Used

- **Programming Language:** Python
- **Libraries:** OpenCV, NumPy, Pandas
- **Algorithm:** Haar Cascade (face detection), LBPH (face recognition)
- **Database:** CSV / SQLite
- **Platform:** Jupyter Notebook / VS Code

8. Work Completed So Far

- Studied face recognition concepts and algorithms
- Collected dataset of facial images
- Implemented face detection using OpenCV
- Began training the recognition model

- Designed basic system workflow
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9. Challenges Faced

- Variations in lighting conditions
 - Difficulty in detecting faces at different angles
 - Handling multiple faces simultaneously
 - Dataset quality affecting accuracy
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10. Future Work

- Improve accuracy using deep learning models (CNN, FaceNet)
 - Add mask detection feature
 - Integrate cloud-based database
 - Develop mobile or web-based interface
 - Enhance real-time performance
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11. Conclusion

The midterm progress of the Attendance System Using Face Recognition shows promising results. The system successfully detects faces and begins the recognition process using machine learning techniques. Although some challenges remain, the project is on track toward developing a fully functional automated attendance system.

This system has the potential to revolutionize attendance tracking by making it more efficient, secure, and user-friendly.